

INTELLIGENT HYBRID PROFESSIONAL NETWORK RECOMMENDATION SYSTEM

Major Project Report

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Submitted to: Unlox Team

Project Area: Machine Learning / Recommendation Systems

1. INTRODUCTION

Professional networking platforms contain large amounts of information about users, including their professional interests, skills, career goals, personality characteristics, location and experience. Identifying professionally compatible users can be difficult when these factors are considered independently.

The Intelligent Hybrid Professional Network Recommendation System addresses this problem by combining multiple profile attributes into a unified recommendation framework.

The system uses Natural Language Processing (NLP) to process unstructured profile information and calculate semantic similarity. It also incorporates MBTI personality compatibility, skill similarity, location compatibility and experience similarity. These components are combined using a weighted hybrid scoring model to produce a compatibility score between 0% and 100%.

The system also incorporates a feedback mechanism. Users can accept or reject recommendations through the Streamlit application. The feedback is used to adjust recommendation weights for subsequent recommendations.

2. PROBLEM STATEMENT

The objective of this project is to develop a Python-based professional networking recommendation system that identifies professionally compatible users using profile information.

The system should process free-text profile information using NLP, calculate semantic similarity, incorporate personality and professional attributes, generate explainable recommendations, and support feedback-based adaptation of recommendation weights.

3. OBJECTIVES

The main objectives of the project are:

1. To generate and validate professional user profiles.
2. To process unstructured profile information using NLP techniques.
3. To calculate semantic similarity using TF-IDF and cosine similarity.
4. To implement a 16×16 MBTI compatibility matrix.
5. To calculate skill compatibility.
6. To calculate location compatibility.
7. To calculate experience compatibility.
8. To combine multiple factors using a hybrid weighted scoring model.
9. To generate explainable recommendations.
10. To provide compatibility score breakdowns.
11. To collect Accept/Reject feedback.
12. To adapt recommendation weights based on feedback.
13. To provide an interactive Streamlit interface displaying the top five recommendations.

4. SYSTEM ARCHITECTURE

The overall system follows the following workflow:

User Profiles → Data Validation → Text Preprocessing → TF-IDF Vectorization → Cosine Similarity → MBTI Compatibility → Skill Matching → Location Matching → Experience Matching → Hybrid Compatibility Score → Top-5 Recommendations → Explainability → User Feedback → Adaptive Weights

The system is divided into the following major components:

5. Dataset

The final dataset contains 100 professional user profiles and 35 attributes.

The attributes include personal information, professional information, skills, career goals, personality information, work preferences and other professional characteristics.

The final dataset validation produced the following results:

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DATASET VALIDATION REPORT

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Total Records	: 100
Total Columns	: 35
Missing Values	: 0
Duplicate Rows	: 0
Duplicate User IDs	: 0
Duplicate Names	: 2
Unique Professions	: 20
Age Range	: 21 - 35
Experience Range	: 0 - 14
Unique Cities	: 15
Unique States	: 13

Figure 1: Dataset Validation

The dataset contains 20 professions with five profiles generated for each profession.

The dataset was validated for missing values, duplicate records and duplicate user identifiers.

6. Exploratory Data Analysis

Additional exploratory analysis was performed on:

- Profession distribution
- MBTI distribution
- Age distribution
- Experience distribution
- Gender distribution
- Preferred work mode
- Industry distribution
- Top skills
- Compatibility scores

- Feedback distribution
- Updated compatibility scores
- City distribution

Select a User

Choose a User ID

U006

User Profile

Name: Shruti Iyer

Profession: Full Stack Developer

Industry: Software Development

City: Bhubaneswar

Experience: 11

MBTI: ENTP

Career Goal: Lead software development teams

Skills: REST APIs, MongoDB, React, CSS, Docker, Python

 Generate Recommendations

Figure 2: Profession Distribution

The validation confirmed that the final user dataset contains no missing values, duplicate rows or duplicate User IDs.

7. NLP-Based Profile Similarity

The recommendation system uses NLP to process unstructured profile information.

The following profile fields are combined for text processing:

- Profession
- Skills
- Career Goal

- Professional Summary
- About Me
- Interests
- Personality Traits
- Personal Values
- Favorite Tech Stack
- Preferred Industry

TF-IDF vectorization is applied with English stop-word removal, a maximum vocabulary size of 500 features and unigram/bigram features.

Cosine similarity is then calculated between the TF-IDF representations of users.

The resulting similarity matrix contains pairwise similarity values for the 100 users and has a dimension of:

100 × 100

```
similarity_matrix.shape
```

```
[100]:
```

```
(100, 100)
```

Figure 3: Similarity Matrix

The matrix is saved in 'similarity_matrix.csv' for use by the recommendation engine.

8. MBTI Compatibility

The system incorporates personality compatibility using a 16 × 16 MBTI compatibility matrix.

	INTJ	INTP	ENTJ	ENTP	INFJ	INFP	ENFJ	ENFP	ISTJ	ISFJ	ESTJ	ESFJ	\
INTJ	100	90	80	70	85	75	65	55	85	70	65	50	
INTP	90	100	70	80	75	85	55	65	75	60	55	40	
ENTJ	80	70	100	90	65	55	85	75	65	50	85	70	
ENTP	70	80	90	100	55	65	75	85	55	40	75	60	
INFJ	85	75	65	55	100	90	80	70	70	85	50	65	
INFP	75	85	55	65	90	100	70	80	60	75	40	55	
ENFJ	65	55	85	75	80	70	100	90	50	65	70	85	
ENFP	55	65	75	85	70	80	90	100	40	55	60	75	
ISTJ	85	75	65	55	70	60	50	40	100	85	80	65	
ISFJ	70	60	50	40	85	75	65	55	85	100	65	80	
ESTJ	65	55	85	75	50	40	70	60	80	65	100	85	
ESFJ	50	40	70	60	65	55	85	75	65	80	85	100	
ISTP	75	85	55	65	60	70	40	50	90	75	70	55	
ISFP	60	70	40	50	75	85	55	65	75	90	55	70	
ESTP	55	65	75	85	40	50	60	70	70	55	90	75	
ESFP	40	50	60	70	55	65	75	85	55	70	75	90	

	ISTP	ISFP	ESTP	ESFP
INTJ	75	60	55	40
INTP	85	70	65	50
ENTJ	55	40	75	60
ENTP	65	50	85	70
INFJ	60	75	40	55
INFP	70	85	50	65
ENFJ	40	55	60	75
ENFP	50	65	70	85
ISTJ	90	75	70	55
ISFJ	75	90	55	70
ESTJ	70	55	90	75
ESFJ	55	70	75	90

	ISTP	ISFP	ESTP	ESFP
ISTP	100	85	80	65
ISFP	85	100	65	80
ESTP	80	65	100	85
ESFP	65	80	85	100

Matrix shape: (16, 16)

Figure 4: MBTI Compatibility Matrix

All 16 MBTI personality types are supported.

The MBTI compatibility scores are project-defined rules used for recommendation scoring and should not be interpreted as scientifically validated measures of professional compatibility.

The scoring begins with a base compatibility value and increases according to shared MBTI dimensions. Users with identical MBTI types receive a compatibility score of 100.

The resulting matrix is used as one component of the hybrid recommendation score.

9. Skill Matching

Skill compatibility is calculated by comparing the skills associated with two users.

The system identifies common skills between users and calculates a percentage-based skill compatibility score.

This allows users with overlapping technical or professional skills to receive higher compatibility contributions.

10. Location Matching

Location compatibility is calculated using the users' cities.

Users located in the same city receive a higher location compatibility score, while users in different cities receive a lower score.

11. Experience Matching

Experience compatibility is calculated using the difference between the users' experience levels.

Users with similar levels of professional experience receive higher experience compatibility scores.

12. Hybrid Recommendation Algorithm

The final compatibility score combines five components:

Text Similarity: 0.50

MBTI Compatibility: 0.20

Skill Matching: 0.15

Location Matching: 0.10

Experience Matching: 0.05

The weighted components are combined to produce a final compatibility score between 0 and 100%.

The candidate users are then sorted by the final compatibility score, excluding the selected user, and the top five users are returned as recommendations.

13. Explainable Recommendations

The system provides explanations in addition to numerical compatibility scores.

The explanation can include:

- Shared skills
- Same MBTI
- Same industry
- Similar career goals
- Same city
- Similar experience level



Figure 6: Compatibility Breakdown

The application also displays a compatibility breakdown consisting of:

- Text Similarity
- MBTI Compatibility
- Skill Match
- Location Match
- Experience Match

This improves the transparency of the recommendation process.

14. Feedback Mechanism

The Streamlit application provides Accept and Reject actions for each recommendation.

Recommendation for Rahul Iyer was accepted!

Figure 7: Recommendation Acceptance

Delimiter:

	User_ID	Recommended_User	Compatibility	Feedback	Updated_Compatibility
1	U001	U044	57.85	1	62.85
2	U001	U036	55.4	1	60.4
3	U001	U002	48.14	0	43.14
4	U001	U068	43.73	1	48.73
5	U001	U021	41.56	1	46.56
6	U002	U001	48.14	1	53.14
7	U002	U044	45.42	1	50.42
8	U002	U075	44.25	1	49.25
9	U002	U036	44.02	1	49.02
10	U002	U038	43.45	0	38.45
11	U003	U074	66.29	0	61.290000000000006
12	U003	U097	59.5	0	54.5
13	U003	U018	57.42	1	62.42
14	U003	U069	57.42	1	62.42
15	U003	U087	39.91	0	34.91
16	U004	U049	66.97	1	71.97
17	U004	U063	62.63	1	67.63
18	U004	U031	60.86	0	55.86
19	U004	U062	59.82	0	54.82
20	U004	U007	48.37	1	53.37
21	U005	U064	72.5	1	77.5
22	U005	U090	71.66	1	76.66

Figure 8: Feedback Details

The application records:

- User ID
- Recommended User ID
- Compatibility
- Feedback

- Updated Compatibility

The feedback is stored in `feedback_detailed.csv`.

The system can therefore retain user interactions and use them for subsequent recommendation-weight adaptation.

15. Synthetic Feedback Evaluation Dataset

A synthetic feedback dataset containing 500 interactions was generated to demonstrate and evaluate the feedback and adaptive-weighting workflow.

A separate feedback-training dataset containing 500 interactions was generated for performance analysis.

The dataset contains:

500 feedback records

100 unique users

5 interactions per user

10 columns

0 missing values

0 duplicate rows

The action distribution was:

Rejected: 370

Accepted: 130

This corresponds to:

Acceptance Rate: 26.0%

Rejection Rate: 74.0%

The average compatibility score in the feedback-training dataset was approximately:

25.93236%

Because this feedback dataset was generated for project evaluation, these values should be interpreted as evaluation results from the generated dataset rather than real-world user behavior.

The project contains separate feedback files for application interaction records and evaluation data. The Streamlit application stores user Accept/Reject interactions in `feedback_detailed.csv`, while `feedback_training.csv` contains **500** generated interactions used for evaluation of the feedback-learning workflow. `feedback.csv` contains the separate recommendation-feedback dataset generated during the notebook workflow.

16. Compatibility vs Feedback Analysis

The relationship between compatibility score and feedback was analysed.

The results show that the acceptance rate increased as the compatibility range increased in the evaluated feedback dataset.

Because the feedback dataset is synthetically generated, this relationship demonstrates the behavior of the evaluation setup and should not be treated as evidence of real-world user preferences.

17. Rule-Based Adaptive Weighting

The recommendation system supports rule-based adaptive weighting based on user feedback. The system adjusts the contribution of recommendation factors according to observed Accept/Reject feedback.

For user U001, the resulting personalized weights were:

 View Personalized Recommendation Weights

The system adjusts these weights based on your previous feedback.

Text Similarity: 0.55

MBTI: 0.20

Skills: 0.15

Location: 0.05

Experience: 0.05

Figure 9: Personalized Recommendation Weights

Compared with the default configuration, the adaptive configuration increases the contribution of text similarity and decreases the contribution of location.

18. Baseline vs Adaptive Recommendations

The baseline configuration was evaluated using the default hybrid recommendation weights:

Text Similarity: 0.50

MBTI: 0.20

Skills: 0.15

Location: 0.10

Experience: 0.05

For U001, the baseline top-five recommendations produced an average compatibility of **49.34%**.

The adaptive configuration used personalized weights generated by the rule-based feedback adaptation mechanism based on U001's observed feedback.

Text Similarity: 0.55

MBTI: 0.20

Skills: 0.15

Location: 0.05

Experience: 0.05

The adaptive top-five recommendations produced an average compatibility of **50.94%**.

Performance Change



System Performance [↔](#)

Total Feedback: 501

Accepted: 282

Rejected: 219

Acceptance Rate: 56.29%

Rejection Rate: 43.71%

Average Compatibility: 55.44%

Figure 10: Streamlit Performance Dashboard

Performance Change:

+1.60 percentage points

This result represents an improvement in average recommendation compatibility for the evaluated U001 top-five recommendations and should not be interpreted as classification accuracy.

19. Streamlit User Interface

Top 5 Professional Recommendations

 **Priya Mehta**

User ID: U031

Profession: Data Scientist

Industry: Information Technology

City: Pune

Career Goal: Develop customer recommendation models

Skills: Python, NumPy, Power BI, Machine Learning, Statistics

Compatibility

70.32%

 **Why this recommendation?** Shared skills: Python, Machine Learning; Same MBTI (ISTJ); Same industry (Information Technology); Similar experience level

Activate Windows
Go to Settings to activate Windows.

> View Compatibility Breakdown

 Accept

 Reject

 **Nikhil Malhotra**

User ID: U049

Profession: Data Scientist

Industry: Information Technology

City: Kochi

Career Goal: Build predictive analytics solutions

Skills: Machine Learning, Power BI, Pandas, Data Visualization, Scikit-learn, Python, Tableau

Compatibility

65.34%

💡 **Why this recommendation?** Shared skills: Data Visualization, Machine Learning, Python; Same industry (Information Technology); Similar career goals; Similar experience level

> View Compatibility Breakdown

👍 Accept

👎 Reject

Bhavya Patel

User ID: U062

Profession: Data Scientist

Industry: Information Technology

City: Chennai

Career Goal: Build predictive analytics solutions

Skills: Power BI, Python, NumPy, Statistics, Pandas

Activate Windows
Go to Settings to activate Windows.

Compatibility

62.84%

💡 **Why this recommendation?** Shared skills: Python, Pandas; Same industry (Information Technology); Similar career goals; Similar experience level

> View Compatibility Breakdown

👍 Accept

👎 Reject

Riya Malhotra

User ID: U063

Profession: Data Scientist

Industry: Information Technology

City: Mumbai

Activate Windows
Go to Settings to activate Windows.

Career Goal: Lead a Data Science team

Skills: NumPy, Pandas, SQL, Tableau, Python

Compatibility

51.39%

 **Why this recommendation?** Shared skills: Python, Pandas; Same industry (Information Technology); Located in Mumbai; Similar experience level

> View Compatibility Breakdown

 Accept

 Reject

 **Kavya Naidu**

User ID: U098

Profession: Machine Learning Engineer

Active
Go to Se

Industry: Information Technology

City: Visakhapatnam

Career Goal: Specialize in Generative AI

Skills: TensorFlow, Pandas, Scikit-learn, Python, NumPy

Compatibility

44.87%

 **Why this recommendation?** Shared skills: Python, Pandas, Scikit-learn; Same industry (Information Technology); Similar experience level

> View Compatibility Breakdown

 Accept

 Reject

Recommendation for Riya Malhotra was accepted!

Figure 11: Top 5 recommendations

The final application is implemented using Streamlit.

The interface provides:

1. User selection.
2. User profile information.
3. Recommendation generation.
4. Top-five recommendations.
5. Compatibility scores.
6. Recommendation explanations.
7. Compatibility breakdown.
8. Personalized recommendation weights.
9. Accept functionality.
10. Reject functionality.
11. Feedback confirmation.
12. System performance information.

The UI provides an interactive demonstration of the complete recommendation workflow.

The Streamlit application serves as the user interface for the recommendation engine; the recommendation logic is implemented separately in `recommendation_engine.py`.

20. Technologies Used

The project uses:

- Python
- Pandas
- NumPy
- Scikit-learn
- TF-IDF
- Cosine Similarity
- Matplotlib
- Faker

- Streamlit

21. Project File Structure

The project contains:

professional_network_recommendation.ipynb

Data generation, validation, NLP processing, similarity calculation, MBTI compatibility, hybrid recommendation development, feedback analysis and performance evaluation.

recommendation_engine.py

Core hybrid recommendation engine, score calculation, explainability, adaptive weight logic and feedback performance evaluation.

app.py

Streamlit-based user interface and feedback interaction.

users.csv

Final 100-user profile dataset.

users_cleaned.csv

cleaned user profile dataset.

similarity_matrix.csv

100 × 100 profile similarity matrix.

feedback_training.csv

500-record feedback dataset used for evaluation.

feedback_detailed.csv

Application-generated feedback records.

22. Limitations

The feedback-training dataset used for evaluation is generated data rather than a large real-world user-interaction dataset.

The MBTI compatibility component is a project-defined scoring mechanism and is not intended to represent scientifically validated personality compatibility.

The adaptive weighting mechanism currently uses a rule-based adjustment based on observed user feedback. More sophisticated machine-learning-based weight optimization could be implemented in future versions.

The NLP component uses TF-IDF and cosine similarity. Transformer-based embeddings could provide richer semantic representations.

23. Future Scope

Future development can include:

1. Transformer-based sentence embeddings.
2. Advanced learning-to-rank recommendation algorithms.
3. Larger real-world feedback datasets.
4. Database-backed feedback storage.
5. User authentication.
6. Cloud deployment.
7. Real-time recommendation updates.
8. More advanced personalization.
9. Continuous online learning.
10. Integration with professional networking platforms.

24. Conclusion

The Intelligent Hybrid Professional Network Recommendation System provides an end-to-end framework for professional compatibility recommendation.

The system combines NLP-based profile similarity, MBTI personality compatibility, skill matching, location matching and experience matching into a hybrid compatibility score.

It also provides explainable recommendations and an interactive Streamlit interface through which users can accept or reject recommendations.

The feedback mechanism enables adaptive recommendation weighting. For **U001**, the average compatibility of the top-five recommendations **increased** from **49.34% using baseline weights** to **50.94% using adaptive weights**, representing an improvement of **1.60 percentage points**.

The project demonstrates the integration of NLP, hybrid recommendation, explainability, personality compatibility, feedback-based personalization and interactive application development in a single professional networking recommendation system.

